Hybrid AC/DC Static Transmission Expansion Planning Using a New Efficient Iterated Greedy Algorithm

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***Abstract*—In recent years, High Voltage Direct Current (HVDC) transmission has become a technically and economically feasible technology, to the point that HVDC networks integrated with AC systems are being developed. However, only a few research works address Transmission Network Expansion Planning (TNEP) considering both AC and DC networks. To develop an AC/DC TNEP approach, precise models that describe the behavior of the transmission lines and their associated converter stations in DC are required. This article proposes a formulation based on a non-convex model and a heuristic optimization method to solve the AC/DC TNEP. The implementation is based on Julia using the Garver AC/DC 6-bus, the IEEE 14-bus AC/DC, and IEEE AC 24-bus test systems.**

***Index Terms*—AC/DC networks, heuristics, HVDC, optimization, transmission expansion planning**

I. Introduction

The TNEP problem aims to determine all the changes in the transmission infrastructure necessary to meet the balance between demand and generation at the lowest cost, within a planning horizon [1]. However, the issues related to TNEP have changed over the years as the energy system has evolved.

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HVDC systems are currently a technically and economically feasible technology. This technology has outstanding characteristics compared to high voltage alternating current (HVAC) transmission such as: *i)* It is mainly preferred for supplying large amounts of power through underground cables, *ii)* it does not present stability problems when transmitting over long distances, and *iii)* HVDC's allows for an asynchronous connection of AC systems [2].

Currently, there are some research works that deal with the use of HVDC networks in the TNEP problem. In [3], a robust multistage TNEP linear model of hybrid high-voltage AC-DC networks is proposed. A hybrid algorithm for AC/DC TNEP using a stochastic simulation method and a mixed-integer linear formulation (MILP), neglecting the converter station model, is presented in [4]. In [5], a dynamic hybrid TNEP linearized model using HVAC and a multiterminal voltage source converter (VSC) based on HVDC alternatives is presented. That model is formulated as a MILP problem and considers the costs of converting existing HVAC transmission lines to HVDC lines. In [6], a model where shunt compensation allows more flexibility to the expansion options is proposed. The TNEP problem is solved using a combination of nonlinear programming and differential evolution (DE). In [7], different relaxations and approximations of the DC grid TNEP problem are presented.

This article focuses on developing a full non-convex mathematical model and its solution method to solve the hybrid AC/DC TNEP. Section II presents the formulation and mathematical model. The optimization approach based on an iterated greedy algorithm is presented in Section III. In Section IV, results using several test systems are presented. The conclusions are summarized in Section V.

II. Formulation

A. DC network model

The resistor r_d, in the DC link, is divided by the number of poles p_d (2 for this work). The power flow in the DC lines is defined as follows:

$$P^{dc,loss} = (r_d/p_d) \cdot (I^{dc}_{ef})^2 \tag{1}$$

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$$P_{ef}^{dc} = U_e^{dc} \cdot p_d \sum_{f \in \Lambda^{dc}} (U_e^{dc} - U_f^{dc})/(r_d) \quad (2)$$

where $P^{dc,loss}$ are the DC line losses, I, U are the DC current and voltage respectively and P_{ef}^{dc} is the power flow in the DC nodes (e and f).

B. AC/DC converter station model

The AC/DC converter station model consists of 1) a transformer with taps and impedance in series, 2) a filter as shunt susceptance, 3) a phase reactor as series impedance, and 4) a power-electronic AC/DC converter. The mathematical models of each element of the converter used in the proposed approach are detailed in [9].

C. AC network

The nodal active and reactive power flows in the AC network are represented in (3) - (4) as follows:

$$P_{ij}^{ac} = U_i \cdot \sum_{j \in \Lambda^{ac}} U_j \cdot (G_{ij} \cdot \cos\theta_{ij} + B_{ij} \cdot \sin\theta_{ij}) \quad (3)$$

$$Q_{ij}^{ac} = U_i \cdot \sum_{j \in \Lambda^{ac}} U_j \cdot (G_{ij} \cdot \sin\theta_{ij} - B_{ij} \cdot \cos\theta_{ij}) \quad (4)$$

where U is the nodal voltage, G is the series conductance, and B is the series susceptance, for $i-j$ branches.

D. TNEP mathematical model

In the proposed formulation, the objective function of the classical TNEP problem is divided into two sub-problems: i) The economic problem of transmission expansion solved by using the proposed heuristic, and ii) the operational problem solved using mathematical programming-based approaches. The AC and DC network equations are solved at each step. In addition, the losses in the elements based on power electronics are neglected [8]. For the TNEP AC/DC this definition is expanded considering the costs of adding DC transmission lines and AC/DC converters. In this way, the economic problem for transmission expansion planning is defined as follows:

$$\min v = \sum_{ij} c_{ij}^{ac} \cdot n_{ij} + \sum_{ef} c_{ef}^{dc} \cdot n_{ef} + \sum_{cv} c_{cv} \cdot n_{cv} + w \quad (5)$$

subject to:

$$0 \le n_{ij} \le \overline{n_{ij}} \quad (6)$$

$$0 \le n_{ef} \le \overline{n_{ef}} \quad (7)$$

$$0 \le n_{cv} \le 2 \cdot n_{ef} \quad (8)$$

where v is the objective function, c_{ij}^{ac} is the cost of n added AC circuits between nodes i and j, c_{ef}^{dc} is the cost of n added DC circuits between nodes e and f, c_{cv} is the cost of n_{cv} added converters, and w represents the load shedding cost for active and reactive power for the operational sub-problem.

The operational sub-problem is formulated as follows:

$$\min w = \sum_{k} (\alpha_1 \cdot R_{P_k} + \alpha_2 \cdot R_{Q_k}) \quad (9)$$

subject to:

$$P_{cv}^{dc} + P_{ef}^{dc} + P_L = 0 \quad (10)$$

$$P_{cv} + P_{ij}^{ac} - P_G + P_L - R_P^{ac} = 0 \quad (11)$$

$$Q_{cv} + Q_{ij}^{ac} - Q_G + Q_L - R_Q^{ac} = 0 \quad (12)$$

$$\underline{P_G} \le P_G \le \overline{P_G} \quad (13)$$

$$\underline{Q_G} \le Q_G \le \overline{Q_G} \quad (14)$$

$$\underline{R_P} \le R_P \le \overline{R_P} \quad (15)$$

$$\underline{R_Q} \le R_Q \le \overline{R_Q} \quad (16)$$

$$\underline{U_i} \le U_i \le \overline{U_i} \quad (17)$$

$$\underline{U_e^{dc}} \le U_i \le \overline{U_e^{dc}} \quad (18)$$

where P and Q are the active and reactive power flow for AC and DC circuits, cv converters, generation (G) and load (L), respectively. R_P and R_Q model the active and reactive load shedding by using fictitious generators connected to the load buses. α_1 and α_2 are penalty coefficients due to the contribution of the fictitious active and reactive power generators, respectively. If $w > 0$, then load shedding is required to make the current transmission topology a feasible one. Normally, α_1 is set to high values to penalize the objective function in case of load shedding. In this work, α_2 is set to zero, because the fictitious generators contribute only with active power. Active and reactive power limits of existing and fictitious generators are defined in (13) - (16), respectively. Voltage limits are defined in (17) - (18).

III. Iterated Greedy Algorithm

The proposed optimization approach (see Algorithm 1) consists of two steps:

1. Initial solution (s): The so-called *Forward* heuristic starts from zero added lines. It consists of adding, one by one, all the candidate transmission circuits (l_i) to the system. For each added circuit, an AC optimal power flow (Eq. (7) - (16)) is performed to verify the feasibility of the current topology, and the resulting w value is passed to Eq. (1). This is performed until the first feasible topology ($w = 0$) is found, yielding the first feasible objective function value v. Figure 1 shows the forward method used to construct the initial solution vector.

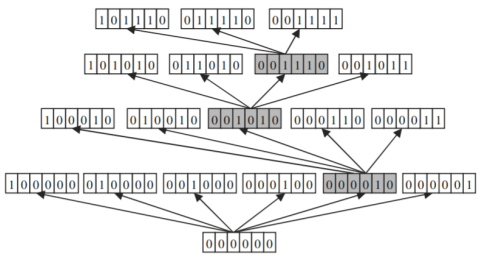


Fig. 1. Forward method - Initial solution vector s [15]

2. Optimization algorithm: *Iterated Greedy* (IG) [16] is a metaheuristic that generates a sequence of solutions using

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two main steps: destruction and (re-)construction. During the destruction step, some solution components are removed from the currently best solution. This is done according to a destruction rate (d), which may be fixed or varying (with a step size of 10%) from a range $[d_{\mathrm{LB}}, d_{\mathrm{UB}}]$. Then *Forward* is applied in order to obtain a new candidate solution. If this new solution is better than the currently best one, the new solution is taken as the new currently best solution. The algorithm stops after m consecutive non-improving iterations and returns the best solution found. In this work, we set $m = 6$.

Algorithm 1 Iterated Greedy (IG) pseudo-code

```
d := d_LB
m := 0
Forward - Initial solution s
for i in n do
    s_i := run_ACOPF(l_i)
end for
s := minimum(s_i)
Iterated Greedy
while m ≤ 6 do
    s^p := DestroyPartially(s, d)
    s' := Reconstruct(s^p)
    if s' is better than s then
        s := s'
        d := d_LB
        m := 0
    else
        d := d + [0(fixed) or 10%]
        m := m + 1
        if d > d_UB then
            d := d_LB
        end if
    end if
end while
output: s, the best solution found
```

IV. Results

To develop the optimization model, the modeling layer JuMP [11] in Julia [10] was used. In addition, PowerModels [12] and PowerModelsACDC [13] were used to solve the AC Optimal Power Flow (AC-OPF). Ipopt [14] was used as solver for the AC-OPF. In this article, AC and AC/DC test systems have been used. For the AC simulations, the Garver 6-bus and the IEEE 24-bus test systems were used. Details of these test systems can be found in [1], [17] and [18]. In addition, modified Garver 6-bus and IEEE 14-bus test systems were used for the AC/DC tests. Full information on these systems can be found in [14]. For all systems, the maximum allowed number of circuits per right of way is five. For each test, 10 experiments, varying the destruction rate, were performed. The results were obtained using Julia parallel processing with 2 and 8 processors, on an Intel i7, 2.80GHz, 16GB RAM PC.

The criteria used to evaluate the results are:

1) Robustness, e.g., the ability to obtain the same best-found solution for as many trials as possible, as measured by the success rate. In this article, a success rate higher than 90% is considered a robust result.
2) Performance, e.g., the required computation time to find the optimal solution. Performance can also be measured by the average number of the objective function evaluations or the average number of iterations required to obtain the best solution.

A. Garver 6-bus system

This system has 6 buses with a total power demand of 720 MW and 152 Mvar, 6 transmission lines and an installed capacity of 1,140 MW. For this system, 15 candidate links are considered for expansion. The proposed scenario for this system allows the generators to deliver a different level of power at any time between its limits.

Table I shows the resulting transmission topology obtained by the proposed approach. The lowest cost found is 160 ($\$ \times 10^3$) which matches the one found in [17], [18], which is considered the optimal solution.

TABLE I
ADDED LINES FOR THE LOWEST COST. GARVER 6-BUS WITH DISPATCHABLE GENERATION

Bus: Start - End	Added lines	Cost per line ($\$ \times 10^3$)
2 - 6	2	30
3 - 5	2	20
4 - 6	2	30
New lines	6	
Total ($\$ \times 10^3$)	160	

B. IEEE 24-bus system

This system has 24 buses, a total power demand of 8,550 MW and 1,740 Mvar, 34 transmission lines and a maximum generation power of 8,866 MW. 41 candidate right-of-ways are considered for expansion. For this system, a scenario with non-dispatchable generation is analyzed.

TABLE II
ADDED LINES FOR THE LOWEST COST. IEEE 24-BUS WITH NON-DISPATCHABLE GENERATION

Bus: Start - End	Added lines	Cost per line ($\$ \times 10^3$)
6 - 10	1	16
7 - 8	2	16
13 - 14	1	62
New lines	4	
Total ($\$ \times 10^3$)	110	

Table II shows the AC circuit additions after applying the proposed approach. The minimum cost found is 110 ($\$ \times 10^3$), which matches the one found in [17].

C. Performance analysis for the AC test systems

Performance results are shown in Table III-A for the Garver AC test system. The solution method found the optimal solution after 2 iterations, for all the trials. The best performance is obtained using a destruction rate of 20%. The lowest average time obtained is 2 seconds with 8 processors.

Table III-B shows the performance of the solution method for the the IEEE 24-bus AC system. For all the destruction rates, the solution method is very robust. However, a 30% destruction rate allows achieving the optimal solution in the shortest time. The lowest average time obtained is 24 seconds with 8 processors.

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TABLE III
PERFORMANCE FOR AC SYSTEMS

A. GARVER 6-BUS SYSTEM: DISPATCHABLE GENERATION, 10 TRIALS										
	2 processors					8 processors				
Destruction rate	Success (%)	Aver. N. Eval	Aver. Iter	Min. Cost. ($\$ \times 10^3$)	Aver. Time (s)	Success (%)	Aver. N. Eval	Aver. Iter	Min. Cost. ($\$ \times 10^3$)	Aver. Time (s)
20%	100	135	2	160	5	100	135	2	160	2
30%	100	150	2	160	6	100	150	2	160	2
40%	100	150	2	160	6	100	150	2	160	2
50%	100	165	2	160	7	100	165	2	160	2
30%-60%	100	150	2	160	6	100	150	2	160	2
B. IEEE 24-BUS SYSTEM: NON-DISPATCHABLE GENERATION, 10 TRIALS										
	2 processors					8 processors				
Destruction rate	Success (%)	Aver. N. Eval	Aver. Iter	Min. Cost. ($\$ \times 10^3$)	Aver. Time (s)	Success (%)	Aver. N. Eval	Aver. Iter	Min. Cost. ($\$ \times 10^3$)	Aver. Time (s)
20%	100	599	6	110	82	100	580	6	110	26
30%	100	554	4	110	75	100	549	4	110	24
40%	100	615	5	110	93	100	676	5	110	29
50%	100	607	5	110	94	100	886	6	110	38
30%-60%	100	574	4	110	83	100	652	5	110	28

D. AC/DC Garver 6-bus system

For this scenario allowing dispatchable generation, there is only one AC candidate transmission path. Meanwhile, the DC candidate's expansion circuits consist of six DC buses, six DC converters (one for each bus), and 11 DC transmission paths interconnecting the six DC buses. The cost of each multiterminal converter at the end of the DC lines is considered.

Table IV shows the lowest cost solution of 22.7 ($\$ \times 10^3$).

TABLE IV
ADDED LINES FOR THE LOWEST COST. GARVER AC/DC 6-BUS WITH DISPATCHABLE GENERATION

	Bus: Start-End	Added lines	Cost per line ($\$ \times 10^3$)	Converter costs ($\$ \times 10^3$)
AC line	5 - 6	1	5	-
DC lines	2 - 6	1	2.4	8.5
	4 - 6	1	2.8	4
New lines		3	10.3	12.5
Total ($\$ \times 10^3$)			22.7	

E. AC/DC 14-bus system

This system has 14 buses with a total load of 777 MW and 220.5 MVar, an installed capacity of 1197 MW and 20 AC transmission lines. As it does not have any DC circuit, it is expected that the proposed approach integrates these circuits into the system. This scenario considers dispatchable generation, 14 DC buses, 14 DC converters, and 213 candidate DC transmission links interconnecting the DC buses.

TABLE V
ADDED LINES FOR THE LOWEST COST. AC/DC 14-BUS SYSTEM WITH DISPATCHABLE GENERATION

	Bus: Start-End	Added lines	Cost per line ($\$ \times 10^3$)	Converter costs ($\$ \times 10^3$)
DC lines	1 - 3	1	1.0	6.4
	1 - 9	1	1.5	3.9
New lines		2	2.5	10.3
Total ($\$ \times 10^3$)			12.8	

Table V presents the final topology for this scenario. Two added DC transmission circuits with the cost of 12.8 ($\$ \times 10^3$) is the solution that was obtained.

F. Performance for AC/DC test systems

The performance results for the first Garver AC/DC hybrid system are shown in Table VI-A. For a 50% destruction rate, the solution method does not find the lowest cost topology. Destruction rates of 20-30-40% present very robust results and higher performance. The best performance is obtained using a 30% destruction rate. The lowest average time of 16 seconds is obtained using 8 processors.

On the other hand, the results for the AC/DC 14-bus system are shown in Table VI-B. Again, the best performance is obtained for a 30% destruction rate. The implemented parallel processing allows obtaining the solution in 104 seconds, which demonstrates the quality of the proposed approach considering the large search space.

G. Solution method convergence

The curves presented in Figure 2 show—in the context of the Garver AC/DC test system—how the optimal solution is obtained with an increasing number of iterations. The convergence to the optimal solution is obtained in less than 15 iterations, using a 30% destruction rate.

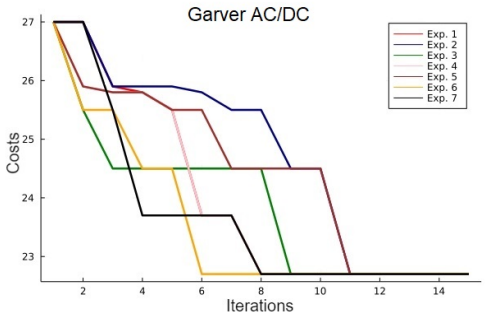


Fig. 2. Solution method convergence for the Garver AC/DC test system.

V. CONCLUSIONS

The continuous technological improvements, the growing need for energy, and the advantages of HVDC technology,

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TABLE VI
RESULTS FOR AC/DC SYSTEMS

A. AC/DC GARVER 6-BUS SYSTEM: DISPATCHABLE GENERATION. 10 EXPERIMENTS

	2 processors					8 processors				
Destruction rate	Success (%)	Aver. N. Eval	Aver. Iter	Min. Cost. ($\$ \times 10^3$)	Aver. Time (s)	Success (%)	Aver. N. Eval	Aver. Iter	Min. Cost. ($\$ \times 10^3$)	Aver. Time (s)
20%	90	230	9	22.7	57	90	228	9	22.7	18
30%	100	206	8	22.7	48	100	218	8	22.7	16
40%	90	218	8	22.7	54	90	230	9	22.7	16
50%	0	419	11	23.7	103	0	404	11	23.7	36
30%-60%	80	260	9	22.7	64	80	280	10	22.7	20

B. AC/DC 14-BUS SYSTEM: DISPATCHABLE GENERATION. 10 EXPERIMENTS

	2 processors					8 processors				
Destruction rate	Success (%)	Aver. N. Eval	Aver. Iter	Min. Cost. ($\$ \times 10^3$)	Aver. Time (min)	Success (%)	Aver. N. Eval	Aver. Iter	Min. Cost. ($\$ \times 10^3$)	Aver. Time (s)
20%	0	2343	9	27.0	12	0	2343	9	27.0	240
30%	100	1150	2	12.8	6.4	100	1324	3	12.8	104
40%	100	1235	2	12.8	6.7	100	1363	3	12.8	145
50%	100	1405	3	12.8	8.2	100	1448	3	12.8	143
30%-60%	100	1491	3	12.8	9.3	100	1406	3	12.8	138

among other factors, directly affect the TNEP problem. These challenges demand the use of more accurate TNEP formulations. This article proposes a mathematical formulation that can be solved using an *Iterated Greedy* metaheuristic. The *Forward* heuristic is used as the construction engine for the candidate solutions. The results indicate rather high robustness and very promising time performance for all the scenarios.

For the AC systems, the expected results were obtained in lower computing times than those obtained in [17] and [18], demonstrating the high performance of the algorithm. For the AC/DC systems, the results are also encouraging. Full success rates were obtained in relatively low calculation times, demonstrating high performance and robustness.

The parameter that most affects the performance of our approach is the destruction rate. A 30% destruction rate presented the best performance, which is the only parameter that must be tuned.

Julia's parallel processing tool reduced drastically the computation time even for the largest search spaces.

The proposed formulation results in a general formulation for the AC/DC static TNEP planning when using heuristic solution methods. If the DC and converter terms are neglected, the classical AC formulation is obtained. More research work is undergoing to apply this approach for even more realistic electric power networks.

VI. ACKNOWLEDGEMENTS

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